



多類別分類演算法與模型驗證方法

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ML classification methods

1. K-nearest neighbor (K-NN) classifier
2. Linear discriminant analysis (LDA) classifier
3. Quadratic discriminant analysis (QDA)
4. Mahalanobis Discriminant Analysis (MDA) classifier
5. Support vector machine (SVM) : a binary classifier
6. Multi-class SVM classifier

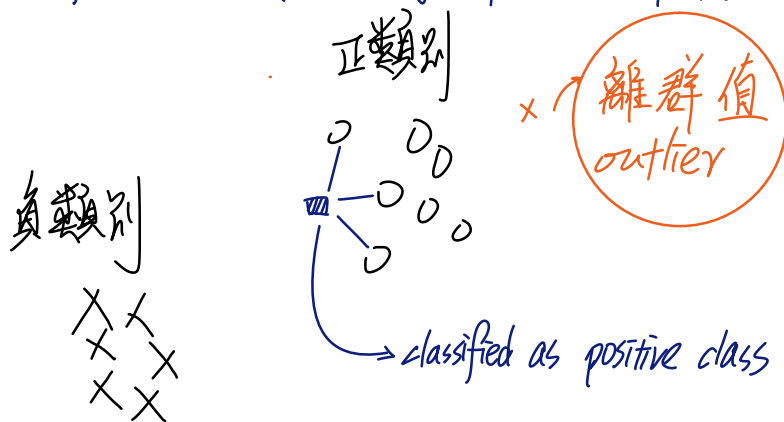


K-NN 解說範例

user defined $\Rightarrow$ need to optimi

④ $\vec{x} \in \mathbb{R}^n$: test data

if $k=3$ 找最接近的 $\Rightarrow$ 第 data $\Rightarrow$ positive



n feature



$x \in \mathbb{R}^n$

Positive: 0

Negative: X

* 若 $k=2$ 可能會出現 tie situation

$\Rightarrow$ 一正 一負

$\Rightarrow$ 盡量用奇數

if $k=1 \Rightarrow$ Nearest Neighbor algorithm

① 优: 快. 易 NN

② 缺: Sensitive to outlier



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K-最近鄰居 (K-nearest neighbor, K-NN)

Given a set of training data $\begin{cases} \text{Training set} : S = \{\mathbf{x}_i \in \mathbb{R}^d\}, i = 1, 2, \dots, N \\ \text{Class label} : y_i \in \{-1, +1\} \end{cases}$ training set size

$\mathbf{x} \in \mathbb{R}^d$, a test data $\notin S$ initialize $K = K$ value (e.g. $K = 1, 3$, or, ...)

Step 1 : Find the K training data points (among the N) that are nearest to $\mathbf{x}$ in the d -dimension space

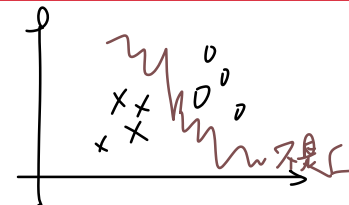
i.e. $\min(K \text{ 個}) \overset{\text{norm}}{\|\mathbf{x}_i - \mathbf{x}\|}, i = 1, 2, \dots, N$

Step 2 : If the majority of the K training data belong to class j^* , then $\mathbf{x} \in j^*$

Remark :

- ① K is the only parameter (user-defined) of the K-NN Algorithm 唯一參數
- ② K can be $1, 2, 3, 4, 5, \dots$ But $K \in \text{odd}$ is better to avoid the "tie" situation
- ③ $K = 1$, K-NN is called nearest-neighbor (NN) algorithm

* 判斷是否是 linear
 $\Rightarrow$ 看 division boundary 是否 线性

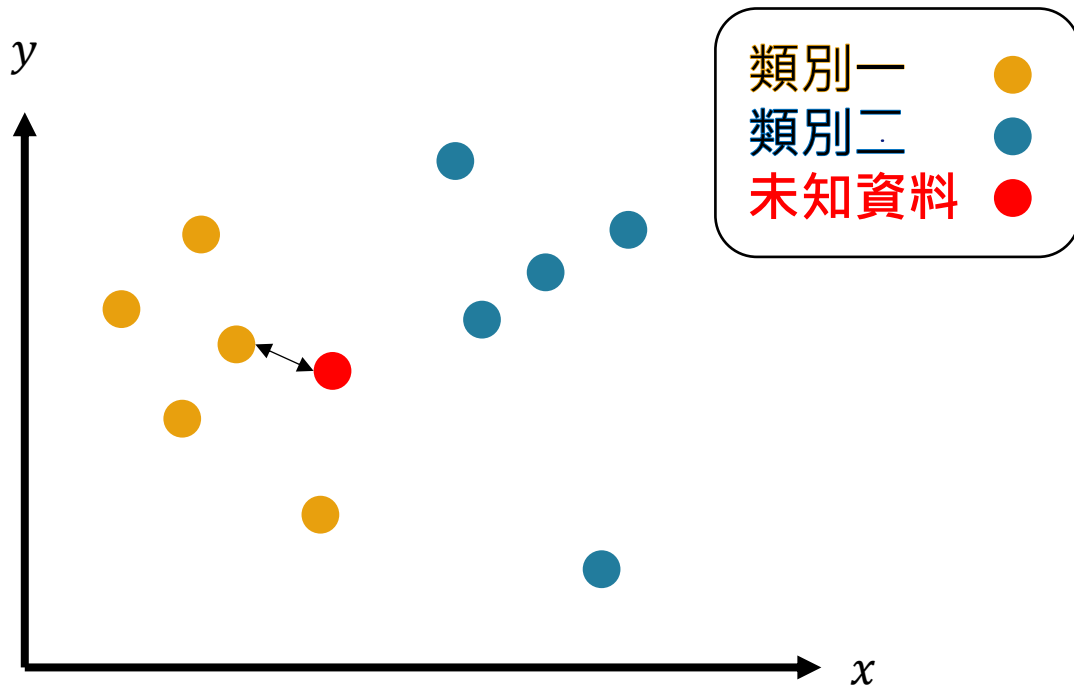




K-NN 範例

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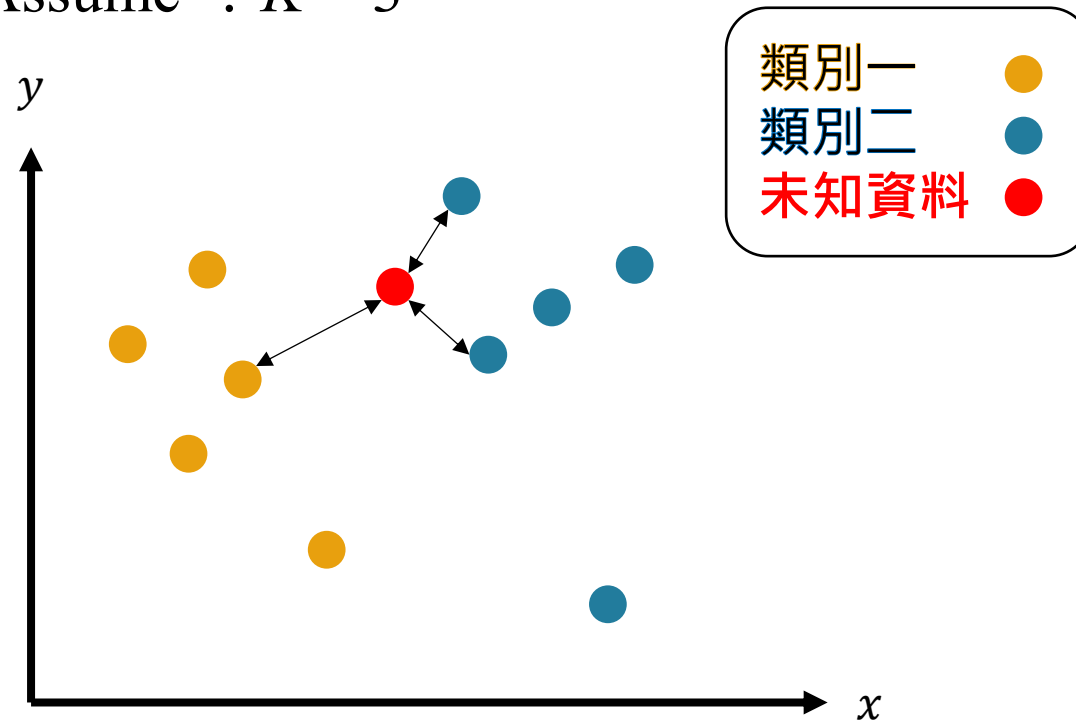
Assume : $K = 1$



Predict

類別一

Assume : $K = 3$



Predict

類別二



混淆矩陣(Confusion matrix)

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➤ 二元分類:

正類別測試資料被分成Positive class, 稱 True Positive(TP) , 反之False Negative (FN)

負類別測試資料被分成Negative class, 稱 True Negative (TN) , 反之False Positive (FP)

Actual \ Predicted	Positive	Negative
Positive	True Positive (TP) 真陽性	False Negative (FN) 偽陰性
Negative	False Positive (FP) 偽陽性	True Negative (TN) 真陰性

Note

① 分類率 = 辨識率(Recognition rate) = 準確率(Accuracy)

② True Positive Rate(TPR) = Sensitivity(敏感性) = $\frac{TP}{TP+FN}$ 判斷陽性準確率

③ True Negative Rate(TNR) = Specificity (特異性) = $\frac{TN}{TN+FP}$ 判斷陰性準確率

◆ 分類率(Classification Rate , CR):

$$CR = \frac{TP+TN}{TP+FN+FP+TN} \times 100\%$$

測試總數

$$= \frac{\text{判斷正確資料數}}{\text{資料總數}} \times 100\%$$

◆ 校正/平衡正、負類別資料數量差異大

$$\text{Balanced CR} = \frac{(TNR+TPR)}{2}$$

✖ 判斷比單看一個好



混淆矩陣範例

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- 假設有50筆測試資料，其中38筆為正類別covid-19(陽性)的資料、12筆為負類別covid-19(陰性)的資料其混淆矩陣如下，其TPR、TNR、分類率(CR)為何？

Actual \ Predicted	Positive	Negative
Positive	TP 36 38	FN 2 error
Negative	FP 4 error	TN 8 12

$$CR = \frac{36 + 8}{50} = 88\%$$

$$TPR = \frac{\# TP}{\# P} = \frac{36}{38} = 95\% \text{ 判陽準}$$

$$TNR = \frac{\# TN}{\# N} = \frac{8}{12} = 67\% \text{ 判陰不準}$$

$$(TPR + TNR) / 2 = 81\%$$

Ex

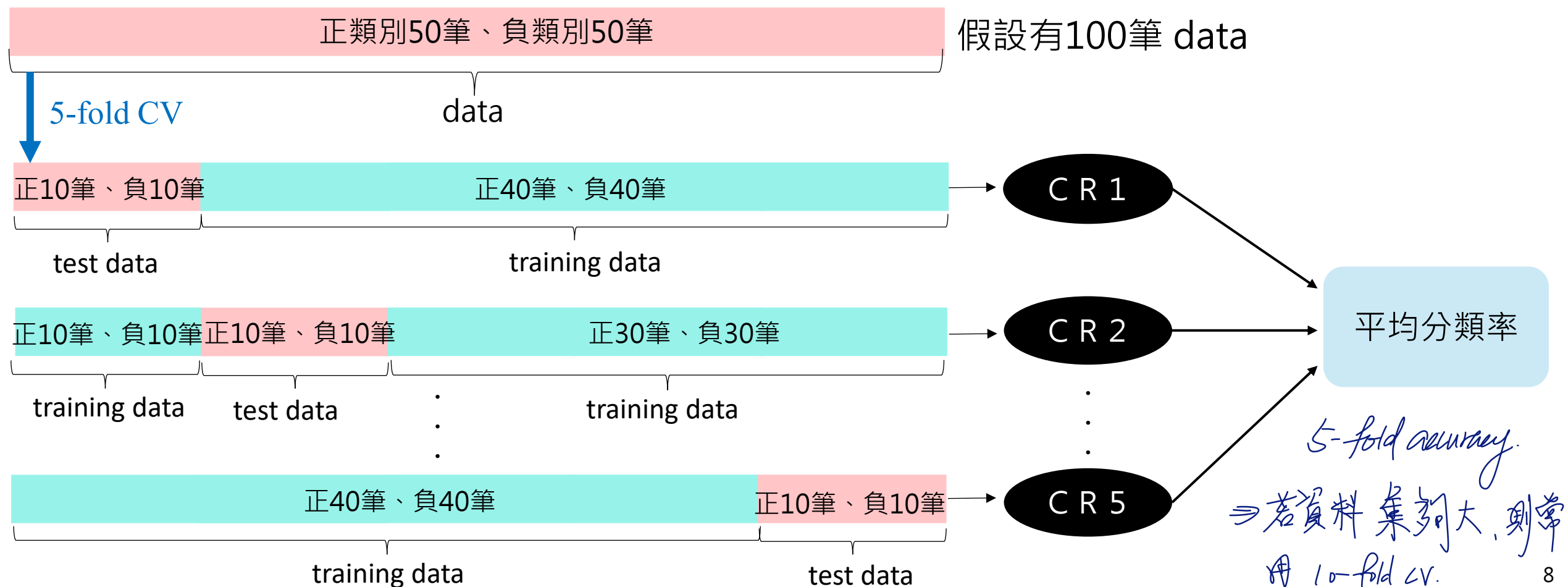


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K -摺交叉驗證 (K -fold cross validation, K -fold CV)

常用 10-Fold

目的：驗證分類器之泛化性 (Generalization) 與是否過度擬合 (Overfitting)





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模型最佳化 → 結合交叉驗證與格子搜尋法

1. 模型最佳化 (model optimization) 亦即將模型的超參數(hyperparameters)最佳化
2. 最佳化 = highest cross validation accuracy

Example : 以 K -NN分類器為例。假設資料集合為 S 。以下說明:

$1\text{-NN} \Rightarrow 80\%$
 $3\text{-NN} \Rightarrow 85\%$
 $5\text{-NN} \Rightarrow 87\%$
 $7\text{-NN} \Rightarrow 83\%$

} 10-fold cv.

格子搜尋法

Algorithm

Initialize k

Step 1: Test 10-fold cv accuracy on T . ($Acc(k)$)

Step 2: change the grid of k ($k = k + 1$)

repeat step 1 until all pre-define grids are tested

Step 3: optimum $k^* = \arg \max_k \text{Acc}(k)$
準確率



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線性鑑別分析(Linear Discriminant Analysis ,LDA)

- 目的：在 $\mathbb{R}^d$ 空間中，找到一個超平面 $H=0$ 去試圖分開兩類別的資料

Def : (Hyperplane in $\mathbb{R}^d$) $H=0$, $\mathbf{x} \in \mathbb{R}^d \rightarrow \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \in \mathbb{R}^d$ d : 維度(dimension)

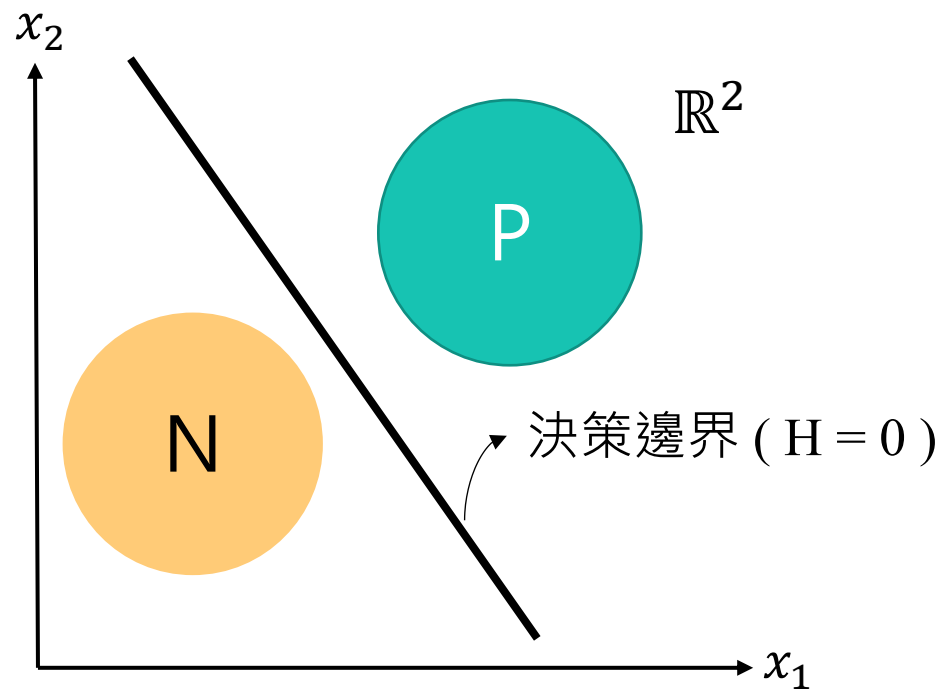
Then $H = 0$

$$\rightarrow w_1x_1 + w_2x_2 + \dots + w_dx_d + b = 0$$

$$\rightarrow [w_1 \ w_2 \ \dots \ w_d] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} + b = 0, \text{ Let } \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \in \mathbb{R}^d$$

$$\rightarrow \mathbf{w}^T \mathbf{x} + \boxed{b} = 0$$

超平面的偏差值(bias)



! LDA是找到超平面的其中一種方法



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LDA分類器的決策函數 (LDA Decision Function)

$$D(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$$

$$\mathbf{w}^T = (\mathbf{m}_1 - \mathbf{m}_2)^T \Sigma^{-1} \Rightarrow \vec{w}^T = 1 \times d$$

$$b = -\frac{1}{2} \left[\underbrace{(\mathbf{m}_1 - \mathbf{m}_2)^T}_{1 \times d} \underbrace{\Sigma^{-1}}_{d \times d} \underbrace{(\mathbf{m}_1 + \mathbf{m}_2)}_{d \times 1} \right]_{\text{Scalar}} - \ln \left(\frac{C_1 P_2}{C_2 P_1} \right)$$

$$\mathcal{S} = \left\{ (\vec{x}_i, y_i) \right\}_{i=1 \dots N}$$

$$\vec{x}_i \in \mathbb{R}^d$$

其中：

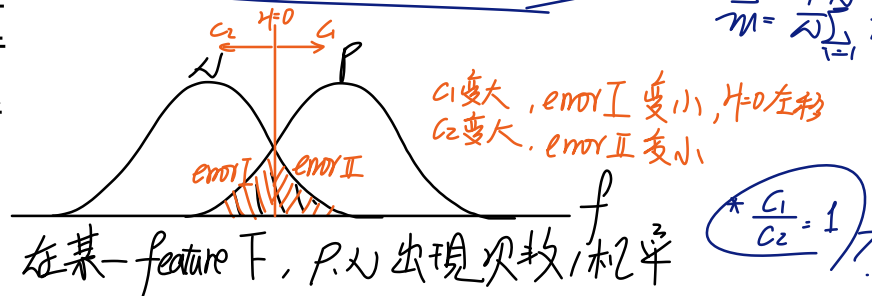
- $\mathbf{w} \in \mathbb{R}^d$ ，為超平面的權重向量
- b 為超平面的偏差值
- $\mathbf{m}_1, \mathbf{m}_2 \in \mathbb{R}^d$ ，為正、負類別訓練資料之平均向量
- $\Sigma \in \mathbb{R}^{d \times d}$ ，為正、負類別訓練資料的共同共變異矩陣
- C_1, C_2 為正、負類別的懲罰權重
- P_1, P_2 為正、負類別的先驗機率

正類別在訓練集中出現機率

$$P_1 = \frac{\#P}{|S|} \quad P_2 = \frac{\#N}{|S|}$$

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^N (\vec{x}_i - \vec{m})(\vec{x}_i - \vec{m})^T$$

$$\vec{m} = \frac{1}{N} \sum_{i=1}^N \vec{x}_i, \in \mathbb{R}^d$$





LDA手算範例: Step 1. 計算 $\mathbf{m}_1, \mathbf{m}_2, P_1, P_2$

假設 $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $\mathbf{x}_4 = \begin{bmatrix} -4 \\ -7 \end{bmatrix}$, $\mathbf{x}_5 = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$, $\mathbf{x}_6 = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$, $\mathbf{x}_1$ 、 $\mathbf{x}_2$ 、 $\mathbf{x}_3$ 為正類別, $\mathbf{x}_4$ 、 $\mathbf{x}_5$ 、 $\mathbf{x}_6$ 為負類別

$$\mathbf{m}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} \mathbf{x}_i \quad \mathbf{m}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} \mathbf{x}_j = \frac{1}{3} [\vec{x}_4 + \vec{x}_5 + \vec{x}_6]$$

其中 $\mathbf{x}_i, \mathbf{x}_j$ 為正、負類別訓練資料, n_1, n_2 為正、負類別訓練資料的數量



LDA手算範例: Step 2. 計算共變異 Σ

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假設 $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $\mathbf{x}_4 = \begin{bmatrix} -4 \\ -7 \end{bmatrix}$, $\mathbf{x}_5 = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$, $\mathbf{x}_6 = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$, $\mathbf{x}_1$ 、 $\mathbf{x}_2$ 、 $\mathbf{x}_3$ 為正類別的訓練資料, $\mathbf{x}_4$ 、 $\mathbf{x}_5$ 、 $\mathbf{x}_6$ 為負類別的訓練資料。

$$\Sigma_1 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (\mathbf{x}_i - \mathbf{m}_1)(\mathbf{x}_i - \mathbf{m}_1)^T =$$

$$\mathbf{m}_1 =$$

其中 Σ_1 為正類別訓練資料的共變異矩陣

$$\Sigma_2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (\mathbf{x}_j - \mathbf{m}_2)(\mathbf{x}_j - \mathbf{m}_2)^T =$$

其中 Σ_2 為負類別訓練資料的共變異矩陣

$$\Sigma = P_1 \Sigma_1 + P_2 \Sigma_2 =$$

其中 $\Sigma \in \mathbb{R}^{d \times d}$, 為正、負類別訓練資料的共同共變異矩陣

LDA手算範例: Step 3. 計算 $\mathbf{w}^T$ 和 b

$$D(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b \quad \begin{cases} \mathbf{w}^T = (\mathbf{m}_1 - \mathbf{m}_2)^T \Sigma^{-1} \\ b = -\frac{1}{2}(\mathbf{m}_1 - \mathbf{m}_2)^T \Sigma^{-1}(\mathbf{m}_1 + \mathbf{m}_2) - \ln\left(\frac{C_1 P_2}{C_2 P_1}\right) \end{cases}$$

其中， C_1 ：正類別的懲罰權重， C_2 ：負類別的懲罰權重 → 使用者自定義參數

$$\text{令 } C_1 = C_2 = 1$$

$$\mathbf{w}^T = [2 - (-3) \quad 3 - (-3)] \begin{bmatrix} -1.6203 & -0.3544 \\ -0.3544 & 0.2025 \end{bmatrix} = [5.9747 \quad -0.5570]$$

$$b = -\frac{1}{2} [2 - (-3) \quad 3 - (-3)] \begin{bmatrix} -1.6203 & -0.3544 \\ -0.3544 & 0.2025 \end{bmatrix} \begin{bmatrix} 2 + (-3) \\ 3 + (-3) \end{bmatrix} - \ln\left(\frac{1 * \frac{3}{6}}{1 * \frac{3}{6}}\right) = 2.9873$$

$$\rightarrow D(\mathbf{x}) = [5.9747 \quad -0.5570]\mathbf{x} + 2.9873$$

$\begin{matrix} > 0 \Rightarrow P \\ < 0 \Rightarrow N \end{matrix}$



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Scatter Plot and Decision Boundary

$$D(\mathbf{x}) = [5.9747 \quad -0.5570]\mathbf{x} + 2.9873$$

Assume P : 失智 N : 正常 $D(\mathbf{x})$ 值: 空間中的距離 $\Rightarrow$ 相似程度

Q: $\bar{x}_a \Rightarrow D(\bar{x}_a) < 0$
 $\bar{x}_b \Rightarrow D(\bar{x}_b) < 0$

$|D(\bar{x}_a)| < |D(\bar{x}_b)| \Rightarrow a$ 更可能失智

N : number of data
 d : dimension ($\mathbf{x} \in \mathbb{R}^d$)

$$\Sigma = \frac{1}{N-1} \sum_i (\mathbf{x}_i - \mathbf{m})(\mathbf{x}_i - \mathbf{m})^T$$

if $N=1$ $d=2$ $\bar{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\Rightarrow \mathbf{x}\mathbf{x}^T = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow \text{rank}=1$ 反矩不存在

$\Rightarrow b$ 無法解

$\Rightarrow$ 當資料數 < 維度 則無法計算

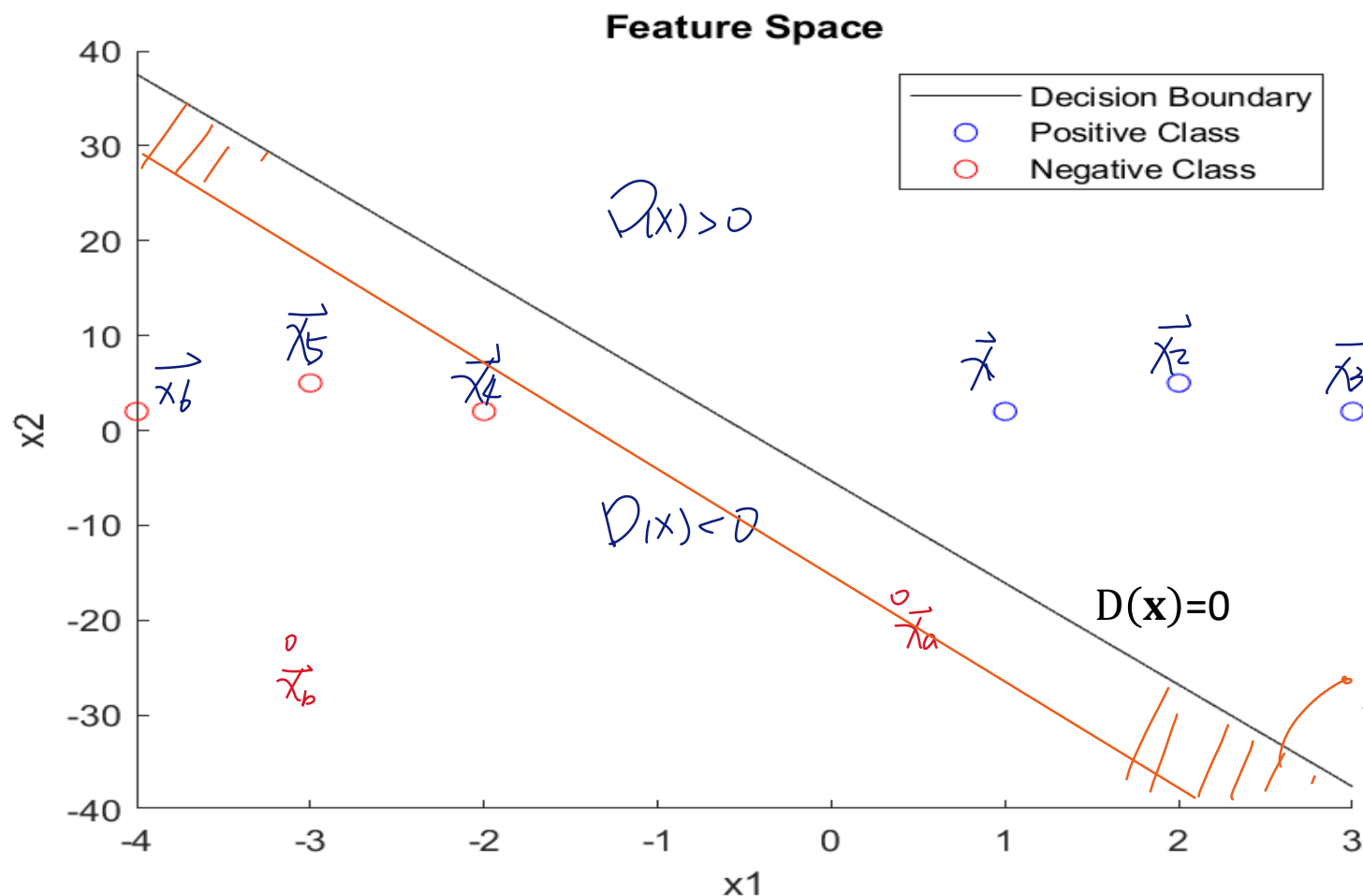
if $N < d$, then Σ^{-1} do not exist

(Small-Sample-Size (SSS) problem)

高維圖效

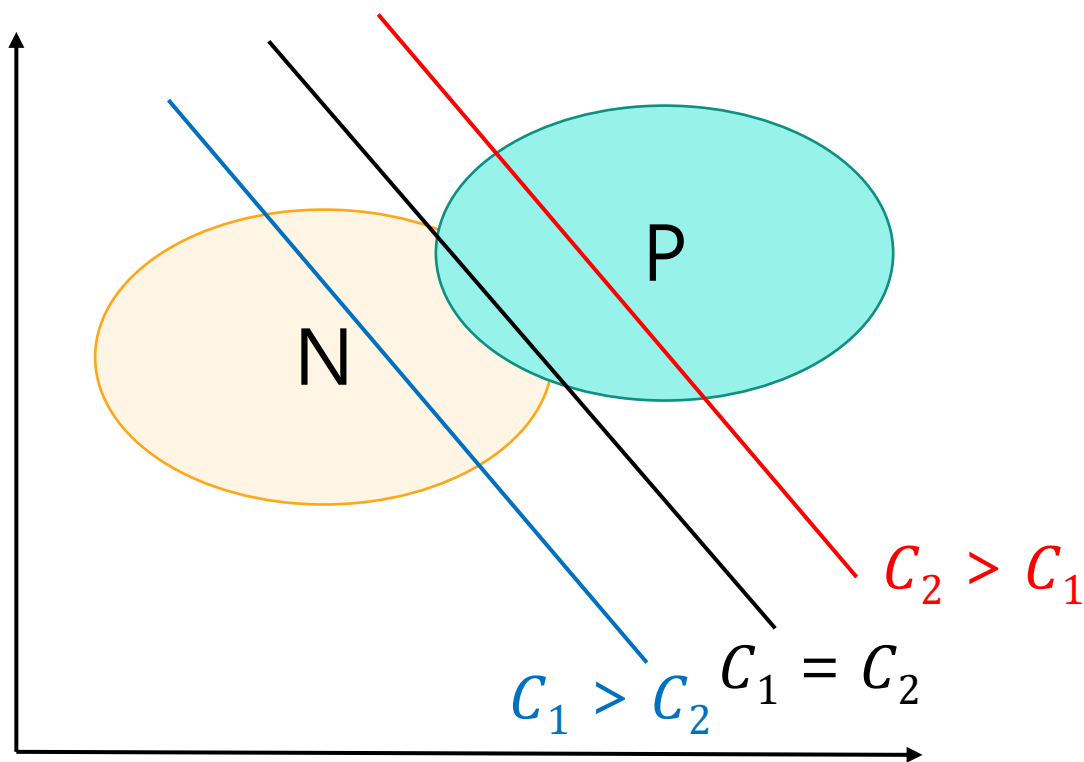
$\epsilon < D(\mathbf{x}) < 0$

解: 若出現則可藉由對角線 + 很小的 ϵ 解決





懲罰權重的設定對decision boundary的影響



Default setting $C_1 = C_2$

Case 1 :

$C_1 > C_2 \rightarrow$ 造成更多的False Positive

Case 2 :

$C_2 > C_1 \rightarrow$ 造成更多的False Negative

重視N (多P判成N)



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QDA classifier

$$\begin{aligned} qdf(\mathbf{x}) &= \frac{1}{2} \mathbf{x}^T (\hat{\Sigma}_1^{-1} - \hat{\Sigma}_2^{-1}) \mathbf{x} + (\hat{\mu}_1^T \hat{\Sigma}_1^{-1} - \hat{\mu}_2^T \hat{\Sigma}_2^{-1}) \mathbf{x} - \frac{1}{2} \ln \left(\frac{|\hat{\Sigma}_1|}{|\hat{\Sigma}_2|} \right) \\ &\quad - \frac{1}{2} (\hat{\mu}_1^T \hat{\Sigma}_1^{-1} \hat{\mu}_1 - \hat{\mu}_2^T \hat{\Sigma}_2^{-1} \hat{\mu}_2) - \ln \left(\frac{C_{12} \pi_2}{C_{21} \pi_1} \right) \end{aligned}$$

$\mathbf{x}$ is the vector to be classified

$\hat{\mu}_1$ and $\hat{\mu}_2$ are the estimated mean vectors of class 1 and 2, respectively

$\hat{\Sigma}$ is the estimated common covariance matrix of the 2 classes

π_1 and π_2 are the a-priori probabilities of the classes, respectively

C_{12} is the cost of false negative and C_{21} is the cost of false positive .



ML classification methods

1. K-nearest neighbor (K-NN) classifier
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MDA classifier

$$mdf(\mathbf{x}) = -\mathbf{x}^T (\hat{\Sigma}_1^{-1} - \hat{\Sigma}_2^{-1}) \mathbf{x} + 2(\hat{\mu}_1^T \hat{\Sigma}_1^{-1} - \hat{\mu}_2^T \hat{\Sigma}_2^{-1}) \mathbf{x} - \hat{\mu}_1^T \hat{\Sigma}_1^{-1} \hat{\mu}_1 - \hat{\mu}_2^T \hat{\Sigma}_2^{-1} \hat{\mu}_2$$

LDA

線性鑑別分析LDA分類器假設類別的資料具有常態分佈 (Normal distribution) 與相同的共變異矩陣 (Covariance matrix)

↓
類別 Cov 相同

QDA

二次鑑別分析QDA分類器不同於線性鑑別分析分類器在於類別的資料不需要有相同的共變異矩陣

MDA

馬氏鑑別分析MDA分類器與二次鑑別分析分類器很相似，不過還是有些例外。第一點，馬氏鑑別分析分類器假設所有類別的先機率完全相同，第二點，此分類基於馬氏距離平方。



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支持向量機 (support vector machine, SVM)

- SVM was proposed by Prof. Vapnik of MIT in 1998.
- SVM is a machine learning method for classification (SVC) and regression (SVR)
- SVM is based on learning principle of structural risk minimization (SRM), which is different from the usual empirical risk minimization (ERM). Hence, SVM can achieve good generalization performance,
- Also, SVM introduces **kernel tricks**. These make SVM superior to other traditional learning machines based on ERM learning strategy, such as the multi-layer perceptions (i.e., the multi-layer neural networks) trained by error back-propagation algorithm.
- Given a hyperparameter, SVM provides a global optimization solution because its solution is obtained by solving a quadratic programming (QP) problem.



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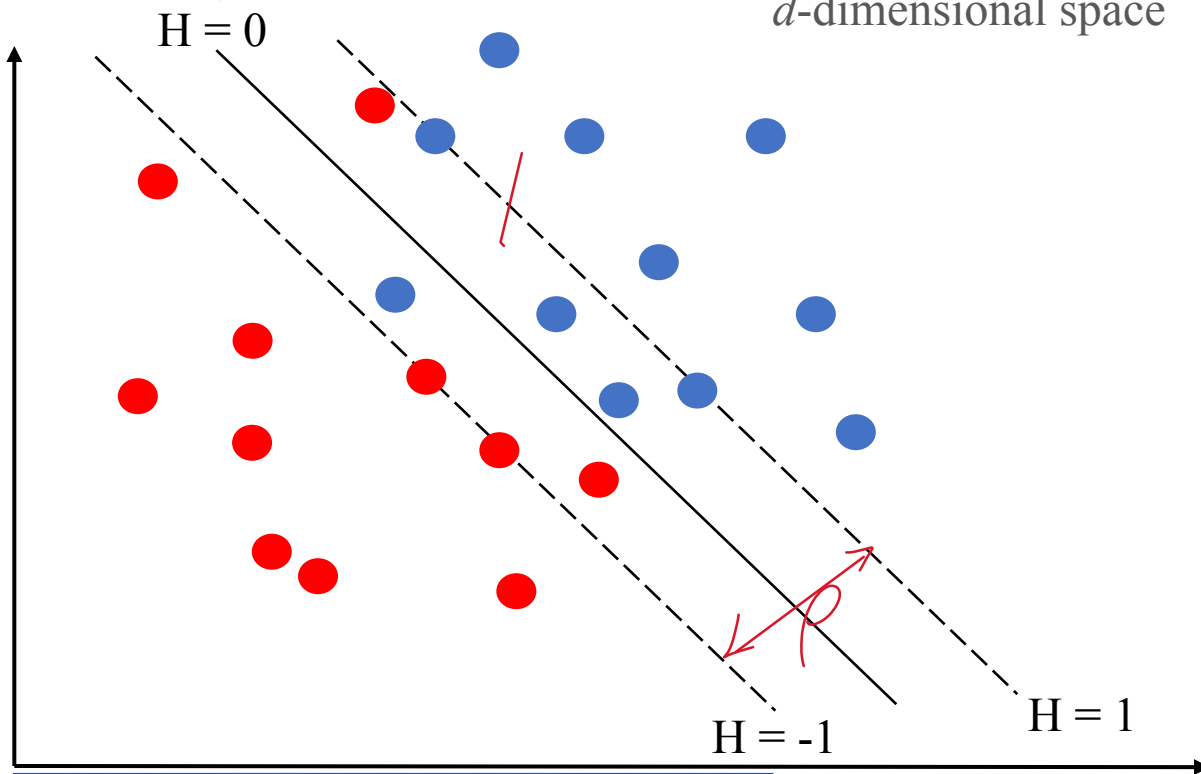
Support Vector Machine (SVM) : 引導解說 (手抄)

Training set: $S = \{(\mathbf{x}_i, y_i)\}$, $i = 1, \dots, N$,

$\mathbf{x}_i \in \mathbb{R}^d$

$$\vec{w}^T \vec{x} + b = 0$$

d -dimensional space



- Positive training data points
- Negative training data points

Idea 1 : maximize the margin of separation between classes

分離邊界

$$\rho = \frac{1}{\|\vec{w}\|}$$

objective : $\max_{(\vec{w}, b)} \rho$

$$\Leftrightarrow \max: \frac{1}{\|\vec{w}\|}$$

$$\Leftrightarrow \min: \frac{\|\vec{w}\|}{1} = \frac{1}{2} \|\vec{w}\|^2 = \frac{1}{2} \vec{w}^T \vec{w}$$

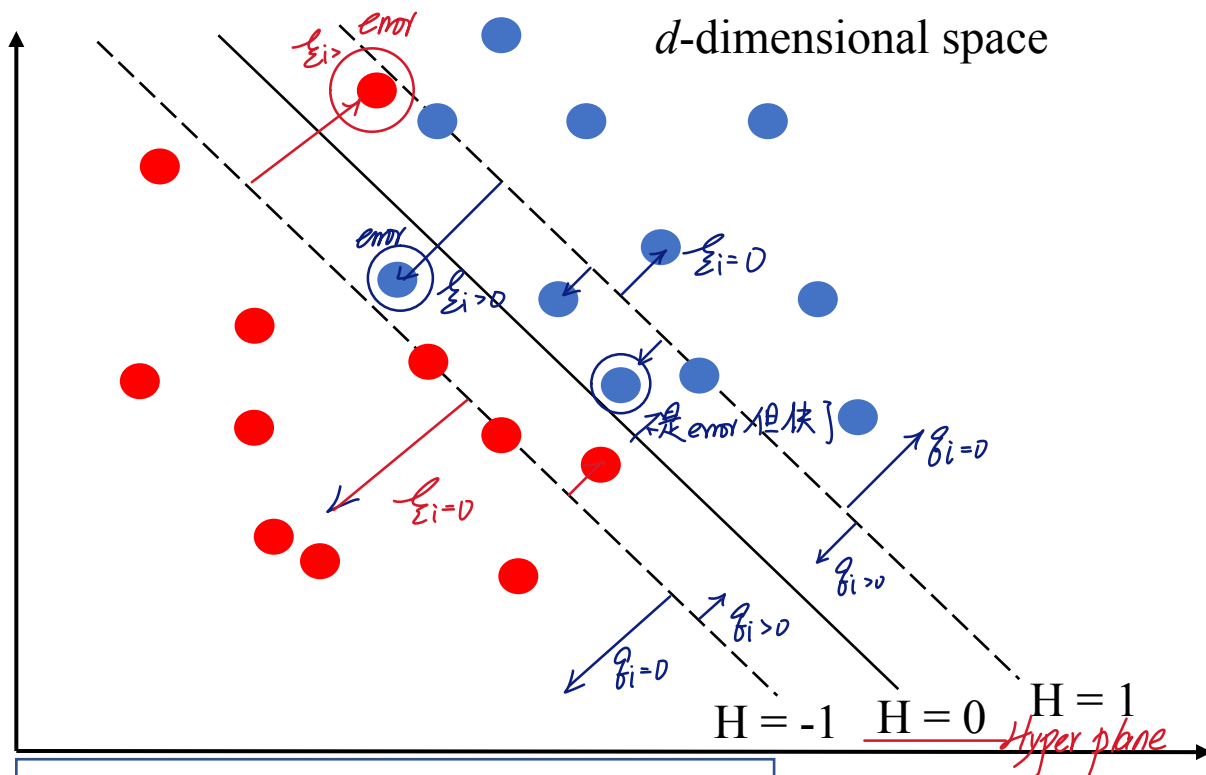


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Training set: $S = \{(\mathbf{x}_i, y_i)\}$, $i = 1, \dots, N$,
 $\mathbf{x}_i \in \mathbb{R}^d$

Idea 2 : minimize the classification errors of the training data points.

Def. $\xi \Rightarrow \Phi$ for $\vec{x}_i \in \text{positive class}$
 if $\vec{x}_i \Rightarrow \bar{L}\vec{x}_i + b > 1 \Rightarrow \xi_i = 0$
 else $\xi_i > 0$
 ② for $\vec{x}_i \in \text{negative class}$
 if $\bar{L}^T \vec{x}_i + b < -1 \Rightarrow \xi_i = 0$
 else $\xi_i > 0$

$$\min \xi_1 + \xi_2 + \dots + \xi_N \Leftrightarrow \min \sum_{i=1}^N \xi_i$$


- Positive training data points
- Negative training data points

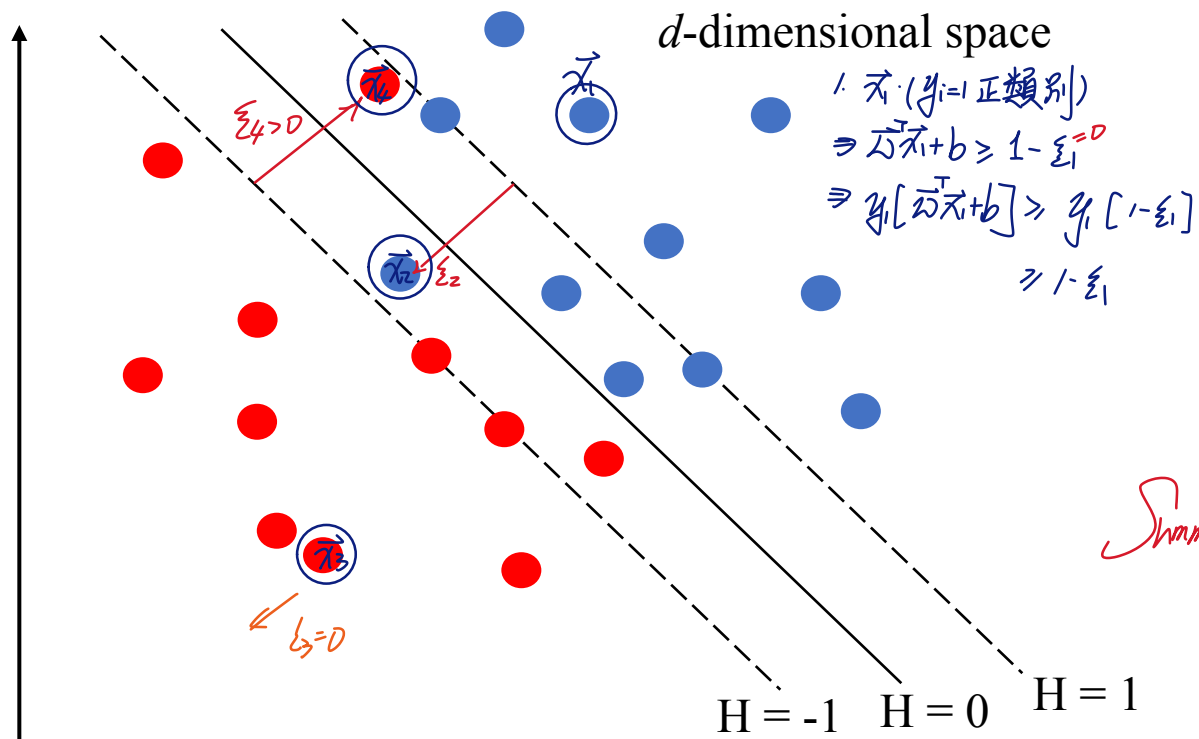


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Support Vector Machine (SVM) : 引導解說 (手抄)

Training set: $S = \{(\mathbf{x}_i, y_i)\}$, $i = 1, \dots, N$,
 $\mathbf{x}_i \in \mathbb{R}^d$

Any constraints that should be satisfied ?



Summary: 所有訓練資料皆須滿足

$$y_i (\bar{\mathbf{w}}^T \mathbf{x}_i + b) \geq 1 - \xi_i, \forall i$$

$$\Rightarrow y_i (\bar{\mathbf{w}}^T \mathbf{x}_i + b) - 1 + \xi_i \geq 0, \forall i$$

Also: ξ_i 鬆弛變數 $\geq 0, \forall i$

若數據集中有大量 $\xi_i > 0$
 $\Rightarrow$ 兩個類別高度重合
 $\Rightarrow$ 特徵選擇不佳

- Positive training data points
- Negative training data points

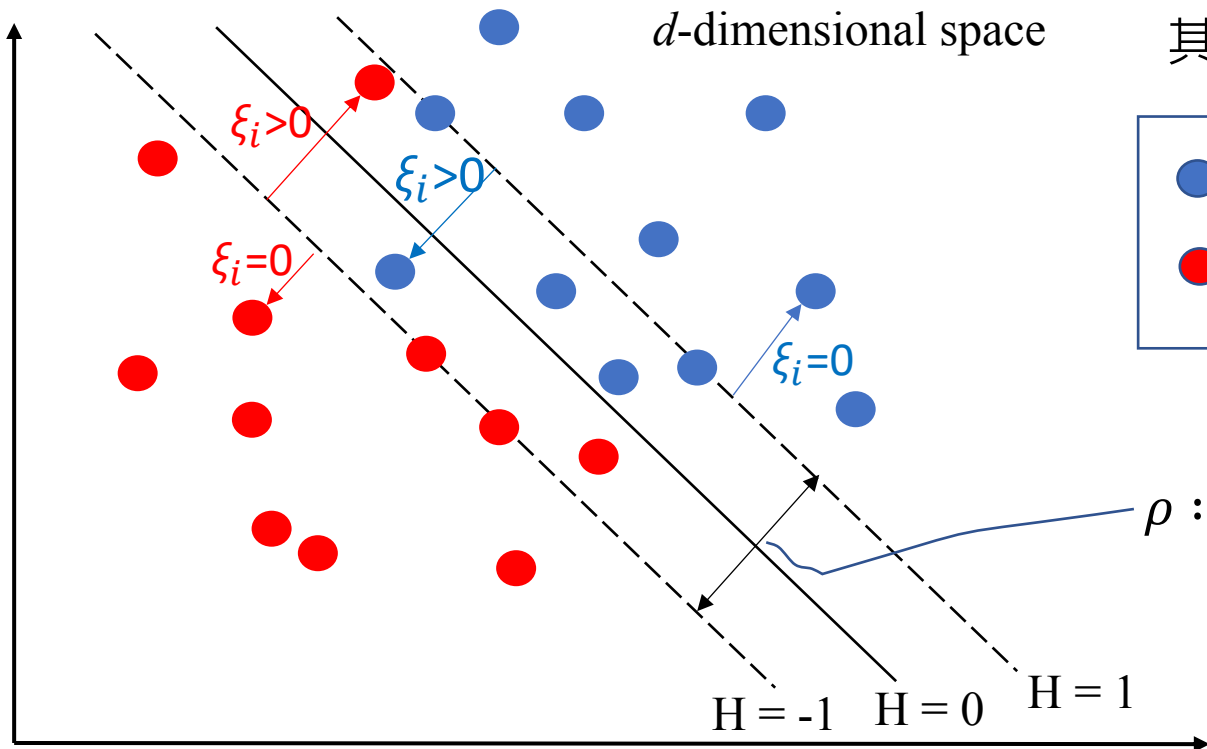


Support Vector Machine (SVM)

- 目的：找到一個最佳分離超平面(Optimal Separating Hyperplane, OSH)將兩類別分離，並滿足下列兩個條件(拘束型最佳化問題)

- 同時完成 {
- ① 最小化訓練資料之分類錯誤 (也就是最大化訓練資料之分類率)
 - ② 最大化分離邊界

$S = \{(\mathbf{x}_i, y_i)\} , i = 1, \dots, N , \mathbf{x}_i \in \mathbb{R}^d$
其中 S 為訓練資料集合



- Positive training data points
- Negative training data points

ρ : margin of separation 分離邊界 $\Rightarrow \max$

ξ_i : slack variables (鬆弛變數) $\Rightarrow \min \sum_i \xi_i$

$H = 0$: separating hyperplane



Linear Support Vector Machine (Linear SVM)

- 最小化誤差 $\rightarrow$ $\underset{(\mathbf{w}, b)}{\text{minimize}} \sum_{i=1}^N \xi_i$
- 最大化分離邊界 $\rightarrow$ $\underset{(\mathbf{w}, b)}{\text{maximize}} \rho = \frac{2}{\|\mathbf{w}\|} \Leftrightarrow \underset{(\mathbf{w}, b)}{\text{minimize}} \frac{\|\mathbf{w}\|^2}{2} \Leftrightarrow \underset{(\mathbf{w}, b)}{\text{minimize}} \frac{1}{2} \|\mathbf{w}\|^2$
 $\Leftrightarrow \underset{(\mathbf{w}, b)}{\text{minimize}} \frac{1}{2} \mathbf{w}^T \mathbf{w}$

Constraints(限制、拘束)

例如： $\xi_i \geq 0, \forall i = 1, \dots, N$

$y_i = +1$

前面兩式乘上 y_i

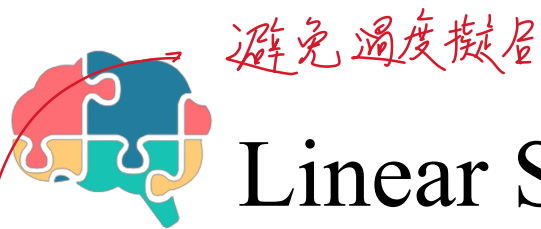
And : If $\mathbf{x}_i \in$ Positive class $\rightarrow \mathbf{w}^T \mathbf{x}_i + b \geq 1 - \xi_i, \forall i \Rightarrow y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i$

$\mathbf{x}_i \in$ Negative class $\rightarrow \mathbf{w}^T \mathbf{x}_i + b \leq -1 + \xi_i, \forall i \Rightarrow y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1 - \xi_i$

$y_i = -1$

其中 $\forall i$ 為 for all i

$\rightarrow$ 所有的訓練資料皆須滿足此 Constraints $\Rightarrow y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 + \xi_i \geq 0$



Linear SVM Formulation

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在原空間 $\mathbb{R}^d$ 找超平面

Given a training set: $S = \{(\mathbf{x}_i, y_i)\}$, $i = 1, \dots, N$, $\mathbf{x}_i \in \mathbb{R}^d$, a linear SVM finds an OSH which can provides maximum margin of separation and minimum training error in the space of original data $\mathbb{R}^d$, which can be formulated as the constrained optimization problem,

多目標最佳化

$$\begin{aligned} &\text{minimize} \\ &(\mathbf{w}, b) \quad \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \end{aligned}$$

目標函數(objection function)

$$\begin{aligned} &\text{subject to} \quad y_i(\mathbf{w}^T \mathbf{x}_i + b) - 1 + \xi_i \geq 0, \forall i = 1, \dots, N \\ &\quad \quad \quad \xi_i \geq 0, \forall i = 1, \dots, N \end{aligned}$$

拘束條件(Constraint)

* C 越大

越重視錯誤率

⇒ 須用 格子搜尋法

try and error

where C : 懲罰權重 (penalty weight) , 使用者自定義的參數 (模型參數)

ξ_i : 鬆弛變數 (slack variables) , 變數

$\mathbf{w}$: weight vector of the OSH , 變數

b : bias of the OSH , 變數

→ 求解 $\mathbf{w}$ 和 b



Limitations of Linear SVM (XOR Problem)

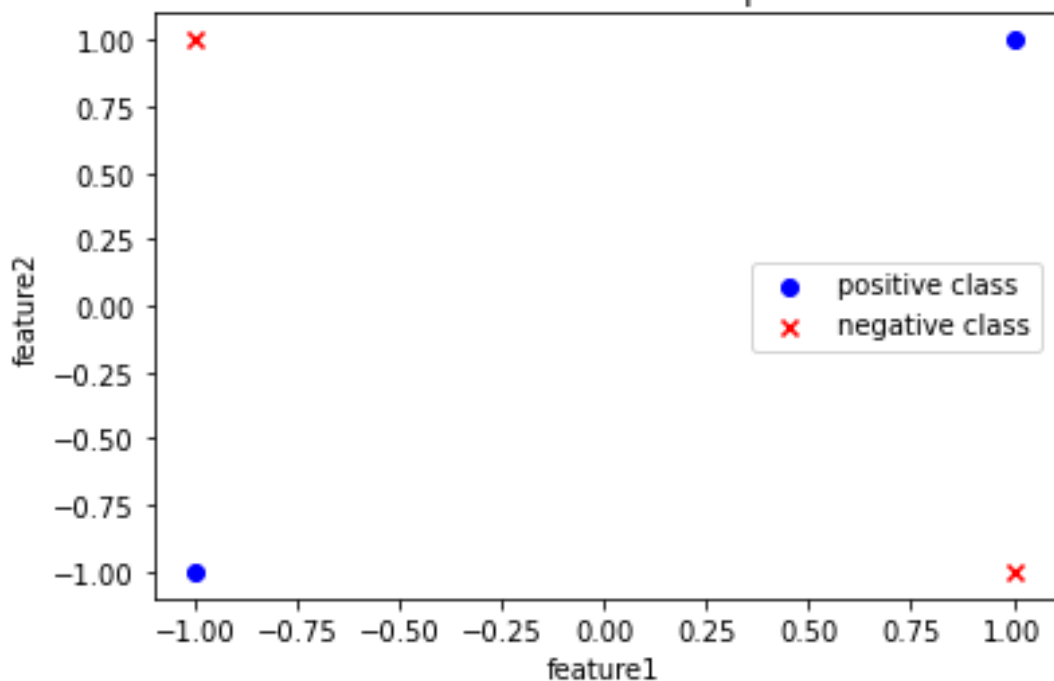
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$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\mathbf{x}_4 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\mathbf{x}_1$ 、 $\mathbf{x}_2$ 為正類別, $\mathbf{x}_3$ 、 $\mathbf{x}_4$ 為負類別

正

負

2-dimension feature space $\mathbb{R}^2$



! 找不到任何超平面分類兩類別

假設 $\mathbf{x} = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \in \mathbb{R}^2$, 定義一個非線性映射函數 ϕ , 將原特徵映射到高維空間

例: $\begin{bmatrix} f_1 \\ f_2 \end{bmatrix}_{\mathbb{R}^2} \xrightarrow{\phi} \begin{bmatrix} f_1 \\ f_2 \\ \sqrt{2}f_1f_2 \end{bmatrix}_{\mathbb{R}^3}$

$$\phi(\mathbf{x}_1) = \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix}$$
$$\phi(\mathbf{x}_3) = \begin{bmatrix} -1 \\ 1 \\ \sqrt{2} \end{bmatrix}$$

$$\phi(\mathbf{x}_2) = \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix}$$
$$\phi(\mathbf{x}_4) = \begin{bmatrix} 1 \\ -1 \\ -\sqrt{2} \end{bmatrix}$$

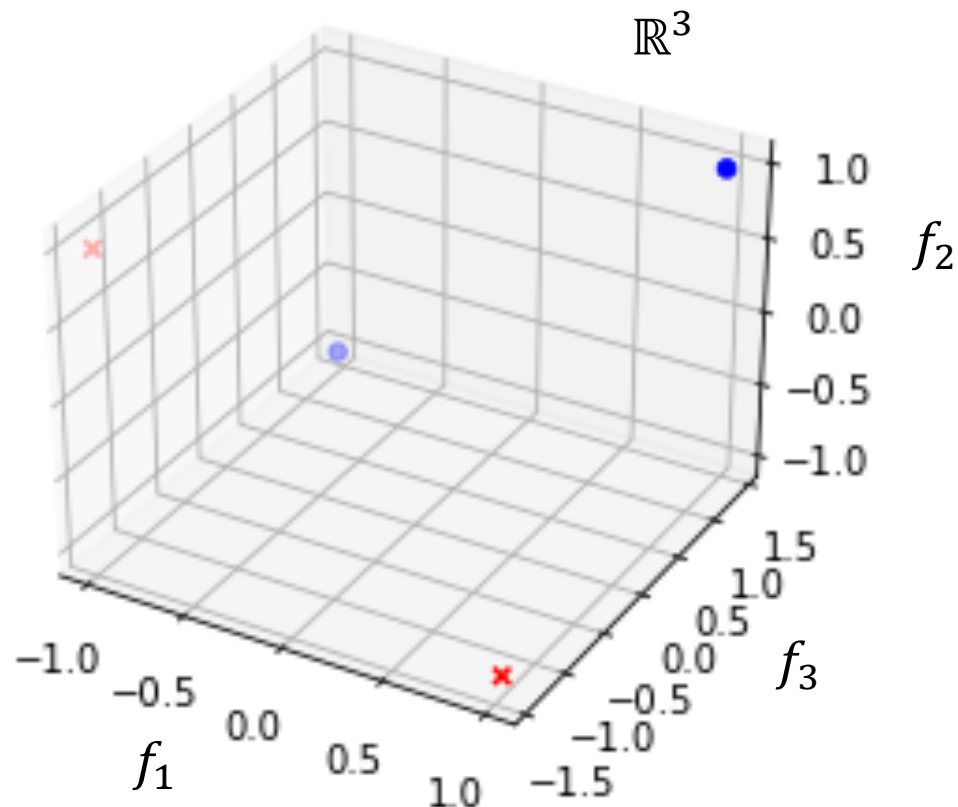
} $\mathbb{R}^3$

在二維空間中是線性不可分離, 但映射至三維後, 成為線性可分離



Limitations of Linear SVM (example: XOR Problem)

$$\phi(\mathbf{x}_1) = \begin{bmatrix} 1 \\ 1 \\ \sqrt{2} \end{bmatrix} \quad \phi(\mathbf{x}_2) = \begin{bmatrix} -1 \\ -1 \\ \sqrt{2} \end{bmatrix} \quad \phi(\mathbf{x}_3) = \begin{bmatrix} -1 \\ 1 \\ \sqrt{2} \end{bmatrix} \quad \phi(\mathbf{x}_4) = \begin{bmatrix} 1 \\ -1 \\ -\sqrt{2} \end{bmatrix}$$



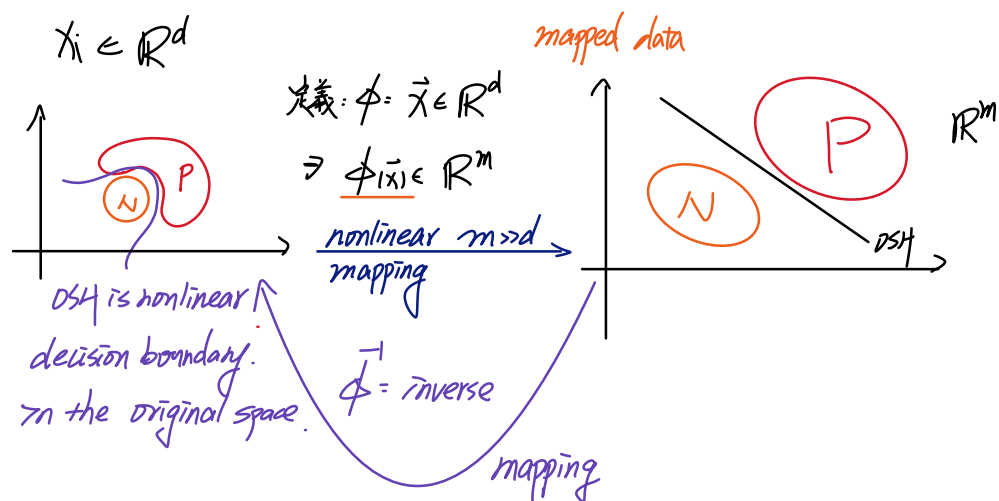
非線性映射到三維的特徵空間後，
這四個資料點變成線性可分離！

⇒ 再找 OSH



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手抄圖例說明SVM映射之概念





Nonlinear SVM

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Why not $\mathbb{R}^m$, 因並非歐氏空間
feature space

- 做法：將原始資料透過非線性映射 ϕ 到高維度空間 F ，使得映射後的資料 $\phi(\mathbf{x})$ ，變成線性可分離，接著在映射後的特徵空間 F 中尋找最佳分離超平面 almost

$$H = \mathbf{w}^T \mathbf{x} + b = 0 \quad \longrightarrow \quad H = \mathbf{w}^T \phi(\mathbf{x}) + b = 0$$

- 目標：求出 $\mathbf{w}$ 和 b $\mathbf{w} \in \mathbb{R}^m$, but ϕ is unknown

$$\left\{ \begin{array}{ll} \text{minimize} & \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \\ (\mathbf{w}, b) & \\ \text{subject to} & y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i \geq 0, \forall i = 1, \dots, N \\ & \xi_i \geq 0, \forall i = 1, \dots, N \end{array} \right\} \quad \text{共有 } 2*N \text{ 個 constraints}$$

稱為 先前問題(Primal problem)



Nonlinear SVM

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- 想法：透過非線性映射(nonlinear mapping) ϕ 將training data映射到更高維的空間(稱為特徵空間 feature space, F)，然後，在從 F 空間中裡尋找最佳分離超平面OSH：

$$H = \langle \mathbf{w}, \phi(\mathbf{x}) \rangle + b = \mathbf{w}^T \phi(\mathbf{x}) + b = 0$$

- 目標 $\rightarrow$ 求解下列的拘束型最佳化問題(constrained optimization problem)來求出 $\mathbf{w}$ 和 b ：

$$\begin{aligned} &\text{minimize} && \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \\ &(\mathbf{w}, b) \end{aligned}$$

$$\begin{aligned} &\text{subject to} && y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i \geq 0, && \forall i = 1, \dots, N \\ &&& \xi_i \geq 0, \forall i = 1, \dots, N \end{aligned}$$

無法直接解 $\Rightarrow$ Lagrangian



求解先前問題

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補充：Consider a constrained optimization problem:

$$\left\{ \begin{array}{ll} \text{Min/Max} & f(x) \\ \text{subject to} & g(x) \geq 0 \\ & h(x) \geq 0 \end{array} \right.$$

where $f(x)$ is the objective function (or cost function), and $g(x) \geq 0$ and $h(x) \geq 0$ are two different inequality constraints. Its Lagrangian Function (L) is defined as

$$L \triangleq f(x) - \alpha g(x) - \beta h(x)$$



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求解SVM的先前問題(primal problem)

primal problem

Step 1 : Construct the Lagrangian

$$\begin{aligned}
 L &= f(x) - \alpha g(x) - \beta h(x) \\
 &= \frac{1}{2} ||\mathbf{w}'||^2 + C \sum_{i=1}^N \xi_i - \\
 &\quad \sum_{i=1}^N \alpha_i [y_i(\mathbf{w}'^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] - \\
 &\quad \sum_{i=1}^N \beta_i \xi_i
 \end{aligned}$$

where α_i and β_i are Lagrange multipliers. 共有2N個 Lagrange multipliers

Step 2 : Take partial differentiation w.r.t the primal variables $\mathbf{w}$, b , and ξ_i , respectively, and set the results to be equal to zero:

$$\frac{\partial L}{\partial \mathbf{w}} = 0; \quad \frac{\partial L}{\partial b} = 0; \quad \frac{\partial L}{\partial \xi_i} = 0$$

$$\begin{aligned}
 \text{minimize} \quad & \frac{1}{2} ||\mathbf{w}'||^2 + C \sum_{i=1}^N \xi_i \\
 \text{subject to} \quad & y_i(\mathbf{w}'^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i \geq 0, \forall i = 1, \dots, N \\
 & \xi_i \geq 0, \forall i = 1, \dots, N
 \end{aligned}$$



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求解SVM的先前問題(primal problem)

$$L = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i - \sum_{i=1}^N \alpha_i [y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] - \sum_{i=1}^N \beta_i \xi_i$$

* $\mathbf{w}$ = 向量

$\Rightarrow$ 值對向量微分

$$\frac{\partial L}{\partial \mathbf{w}} = 0 : \mathcal{L} = \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_i \xi_i - \sum_i \alpha_i [y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] - \sum_i \beta_i \xi_i$$

$$\Rightarrow \mathcal{L}'_{\mathbf{w}} = \mathbf{w} + 0 - \sum_i \alpha_i y_i \phi(\mathbf{x}_i) = 0$$

$$\Rightarrow \mathbf{w}^* = \sum_i \alpha_i y_i \phi(\mathbf{x}_i) \text{ is spanned by the mapped training data set}$$

$\Rightarrow$ SVM 的 opt. weighted vector

= Linear. combine the mapped training data



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求解SVM的先前問題(primal problem)

$$L = \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i - \sum_{i=1}^N \alpha_i [y_i (\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] - \sum_{i=1}^N \beta_i \xi_i$$

$$\frac{\partial L}{\partial b} = 0 : \quad \mathcal{L}'_b = 0 + 0 + 0 - \sum_i \alpha_i y_i = 0$$

$$\Rightarrow \sum_i \alpha_i y_i = 0$$

$$\frac{\partial L}{\partial \xi_i} = 0 : \quad \mathcal{L}'_{\xi} = \overset{*}{0 + C - \alpha_i + \beta_i} = 0$$

$$C = \alpha_i - \beta_i$$

$$\begin{aligned} f &= x_1 + x_2 + x_3 \\ \frac{\partial f}{\partial x_2} &= 1 \end{aligned}$$



求解先前問題

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$$\frac{\partial L}{\partial \mathbf{w}} = 0 \quad \rightarrow \quad \mathbf{w} - \sum_{i=1}^N \alpha_i y_i \phi(\mathbf{x}_i) = 0 \quad \rightarrow \quad \mathbf{w} = \sum_{i=1}^N \alpha_i y_i \phi(\mathbf{x}_i) \quad \text{————— 1}$$

$$\frac{\partial L}{\partial b} = 0 \quad \rightarrow \quad \sum_{i=1}^N \alpha_i y_i = 0 \quad \text{————— 2}$$

$$\frac{\partial L}{\partial \xi_i} = 0 \quad \rightarrow \quad C = \alpha_i + \beta_i \quad \text{————— 3}$$

$$\begin{aligned} & \times \alpha_i, \beta_i \geq 0 \\ & \Rightarrow \beta_i = C - \alpha_i \geq 0 \\ & \Rightarrow C \geq \alpha_i \end{aligned}$$



求解先前問題

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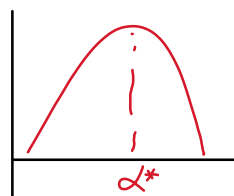
Step 3 : Substitute ① ② ③ back into the Lagrangian function L

$$\begin{aligned}
 L &= \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i - \sum_{i=1}^N \alpha_i [y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] - \sum_{i=1}^N \beta_i \xi_i \\
 &= \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j \phi^T(\mathbf{x}_i) \phi(\mathbf{x}_j) - \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j \phi^T(\mathbf{x}_i) \phi(\mathbf{x}_j) - \sum_{i=1}^N \alpha_i y_i b + \sum_i \alpha_i \\
 &= \sum_i \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j \check{y}_i \check{y}_j \check{\phi}^T(\mathbf{x}_i) \check{\phi}(\mathbf{x}_j) \text{ Scalar.}
 \end{aligned}$$

可觀察到：#N: training dot

1. 所有的先前變數的不見
2. 只剩下變數 $\alpha_1 \dots \alpha_N$ 要求解

$L = L(\alpha_1, \dots, \alpha_N)$ concave function.





求解先前問題之對偶問題(dual problem)

解決下列的對偶問題(Dual problem)即可解出Lagrange multipliers ($\alpha_i \dots \alpha_N$) :

$$\text{Maximize}_{(\alpha_i \dots \alpha_N)} \quad \sum_i^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j \phi^T(\mathbf{x}_i) \phi(\mathbf{x}_j)$$

$$\text{subject to} \quad \sum_{i=1}^N \alpha_i y_i = 0$$

$$0 \leq \alpha_i \leq C, \quad \forall i = 1, \dots, N$$

Note :

- 需要透過程式函數去解Dual problem去得到 $\alpha_i \dots \alpha_N$
- b 無法直接由Dual problem得到最佳解，需要進一步利用Kuhn-Tucker condition to 求解最佳的 b 值

then 求出

$$L^* = \sum_i \alpha_i y_i \phi^T(\mathbf{x}_i)$$



利用Kuhn-Tucker (KT) conditions 求 b 值

定義：若problem為

$$\begin{cases} \text{Min/Max} & f(x) \\ \text{subject to} & g(x) \geq 0 \\ & h(x) \geq 0 \end{cases}$$

則其KT condition 為

$$\begin{cases} \alpha g(x) = 0 \\ \beta h(x) = 0 \end{cases}$$

兩項相乘，一次只能其中一個為0
例如： $\alpha g(x) = 0 \rightarrow \alpha = 0$ or $g(x) = 0$

i.e. 若 $\alpha g(x) = 0$ ；若 $\alpha = 0$ 則 $g(x) > 0$ ；若 $g(x) = 0$ 則 $\alpha > 0$

KT Condition利用Kuhn-Tucker (KT) conditions 求 b 值

$$\underset{(\mathbf{w}, b)}{\text{minimize}} \quad \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i$$

$$\text{subject to} \quad y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i \geq 0, \forall i = 1, \dots, N$$
$$\xi_i \geq 0, \forall i = 1, \dots, N$$

- SVM's KT conditions:

$$\begin{cases} \alpha_i \times [y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] = 0, \forall i \\ \beta_i \times \xi_i = 0, \forall i \end{cases}$$

There are three possible cases for α_i :

- ① $\alpha_i = 0$
- ② $\alpha_i = C$
- ③ $0 < \alpha_i < C$



利用Kuhn-Tucker (KT) condition 求 b 值(^{if $\alpha_i = 0$} Case 1)

Case 1: 若 $\alpha_i = 0 \rightarrow C - \alpha_i > 0 \rightarrow \xi_i = 0$

代表第 i 筆training data在feature space F
中位於margin外(不包含 $H = +1$ or $H = -1$)

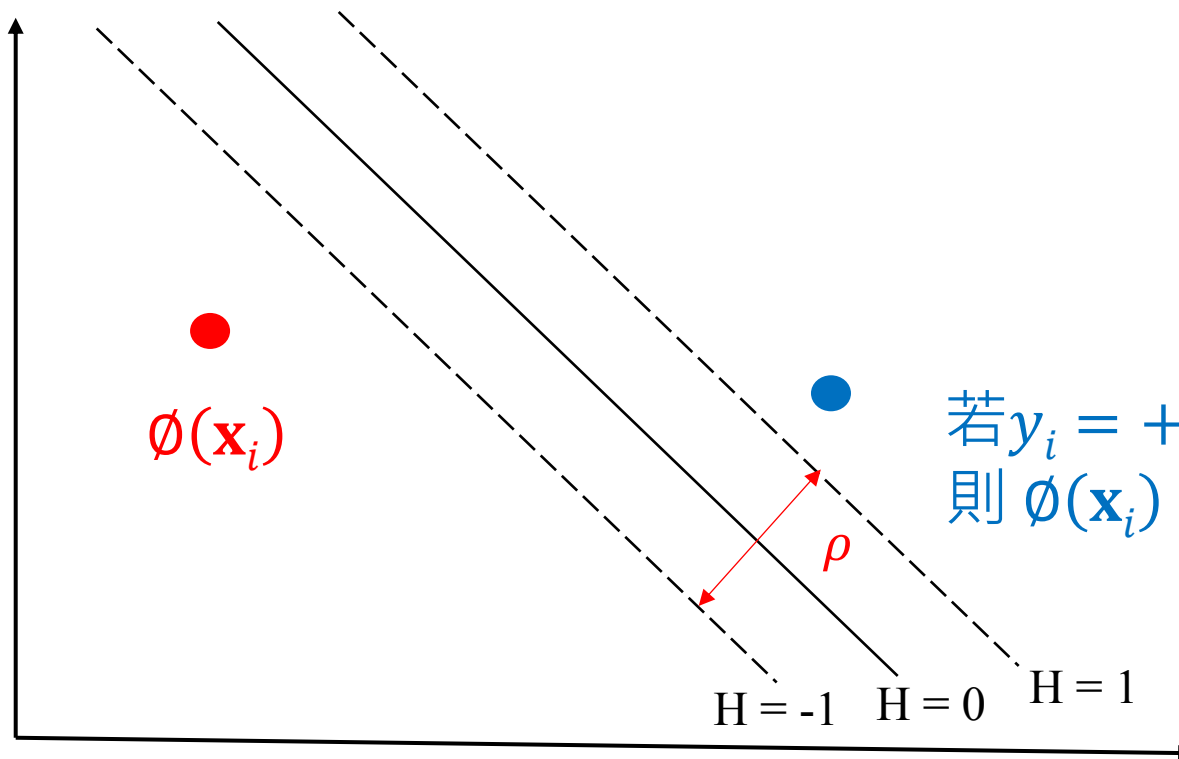
* 由 $\xi_i = 0$ 可知其分類正確
 $\Rightarrow$ 且分類拉的夠遠

若 $y_i = -1$ 且 $\alpha_i = 0$
則 $\phi(\mathbf{x}_i)$ 在 $H = -1$ 的
下方

KT conditions:

$$\alpha_i \times [y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] = 0, \forall i$$

$$\beta_i \times \xi_i = 0, \forall i$$



若 $y_i = +1$ 且 $\alpha_i = 0$
則 $\phi(\mathbf{x}_i)$ 在 $H = 1$ 的上方



利用Kuhn-Tucker (KT) condition 求 b 值(Case 2)

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$$\xi_i = C - \alpha_i > 0 \quad \beta_i \neq 0 \Rightarrow \xi_i = 0$$

Case 2 : 若第 i 筆training data的 α_i 值
為 $0 < \alpha_i < C \Rightarrow \because \alpha_i \neq 0, \therefore \xi_i = 0$

KT conditions :

$$\alpha_i \times [y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] = 0, \forall i$$

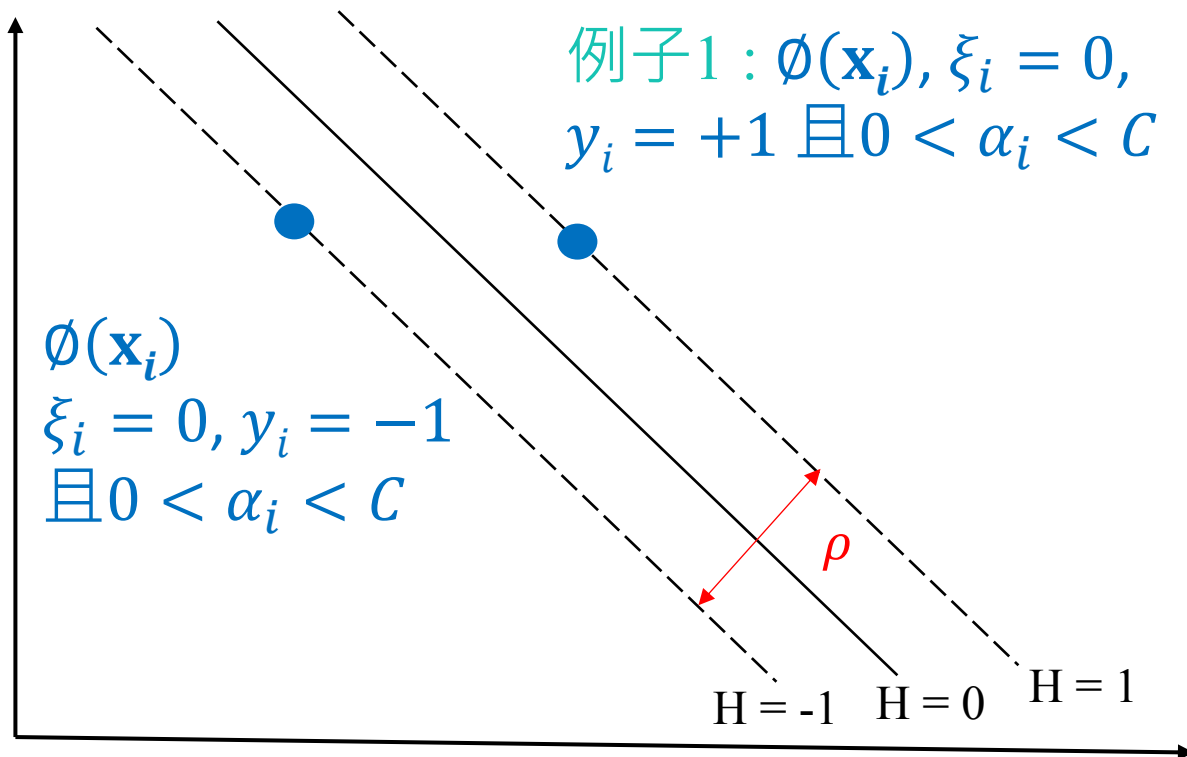
$$\beta_i \times \xi_i = 0, \forall i$$

例子1 : $\phi(\mathbf{x}_i), \xi_i = 0,$
 $y_i = +1$ 且 $0 < \alpha_i < C$

$$y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i = 0$$

$$\Rightarrow y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) = 1 \Rightarrow \begin{cases} y_i = 1, \alpha_i = 1 \\ y_i = -1, \alpha_i = -1 \end{cases}$$

例子2 : $\phi(\mathbf{x}_i)$
 $\xi_i = 0, y_i = -1$
且 $0 < \alpha_i < C$



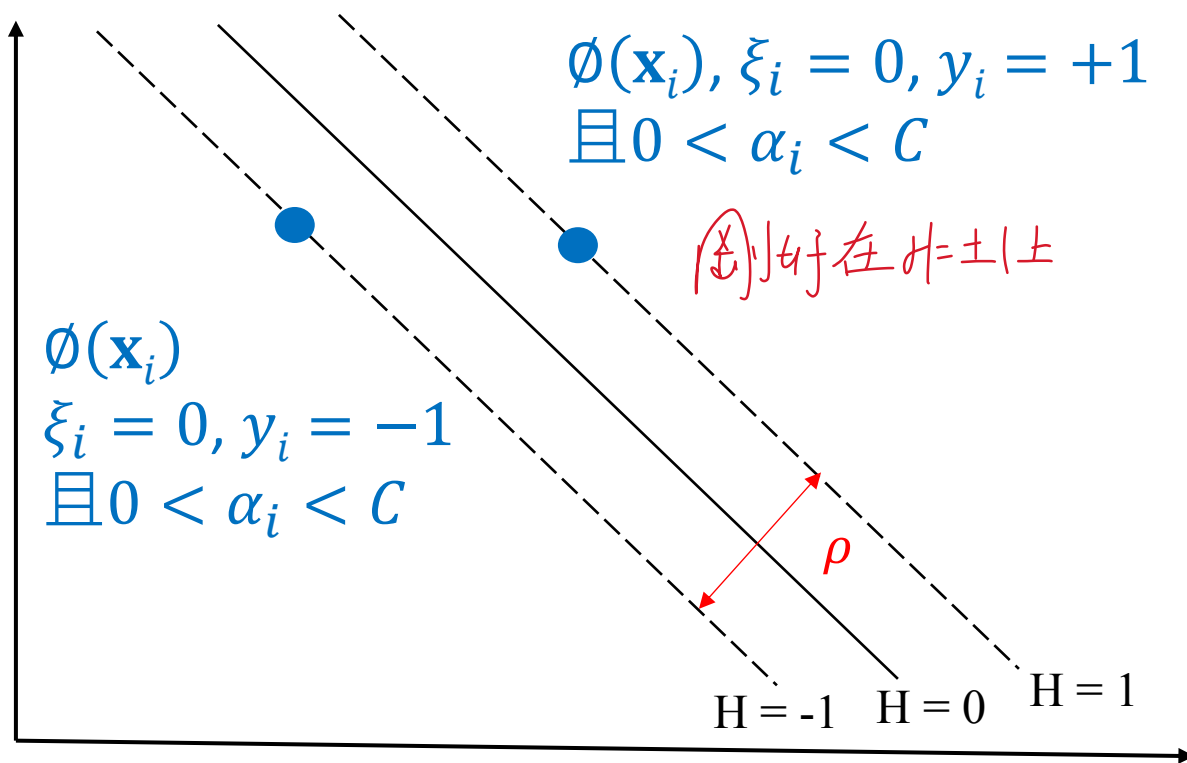
例子1 : 若 $y_i = 1$ ，則 $\mathbf{w}^T \phi(\mathbf{x}_i) + b = 1$ ，代表這筆正類別訓練資料 $\phi(\mathbf{x}_i)$ 剛好在 $H = 1$ 上

例子2 : 若 $y_i = -1$ ，則 $\mathbf{w}^T \phi(\mathbf{x}_i) + b = -1$ ，代表這筆負類別訓練資料 $\phi(\mathbf{x}_i)$ 剛好在 $H = -1$ 上



利用Kuhn-Tucker (KT) condition 求 b 值(Case 2)

Case 2 : 若第 i 筆training data的 α_i 值為
 $0 < \alpha_i < C \Rightarrow \because \alpha_i \neq 0, \therefore \xi_i = 0$



因此，可取一筆 $0 < \alpha_i < C$ 的training data計算 b 的最佳值：

$$\begin{aligned} b^* &= \frac{1}{y_i} - \mathbf{w}^T \phi(\mathbf{x}_i) \\ &= \frac{1}{y_i} - \sum_{j=1}^N \alpha_j y_j \phi^T(\mathbf{x}_j) \phi(\mathbf{x}_i) \end{aligned}$$

where ϕ is still unknown

$$y_i (\bar{\mathbf{w}}^T \phi(\mathbf{x}_i) + b) = 1$$

$$\Rightarrow \bar{\mathbf{w}}^T \phi(\mathbf{x}_i) + b = \frac{1}{y_i}$$

$$\begin{aligned} \Rightarrow b^* &= \frac{1}{y_i} - \bar{\mathbf{w}}^T \phi(\mathbf{x}_i), \quad 0 < \alpha_i < C \\ &= \frac{1}{y_i} - \sum_j \alpha_j y_j \phi^T(\mathbf{x}_j) \phi(\mathbf{x}_i) \end{aligned}$$



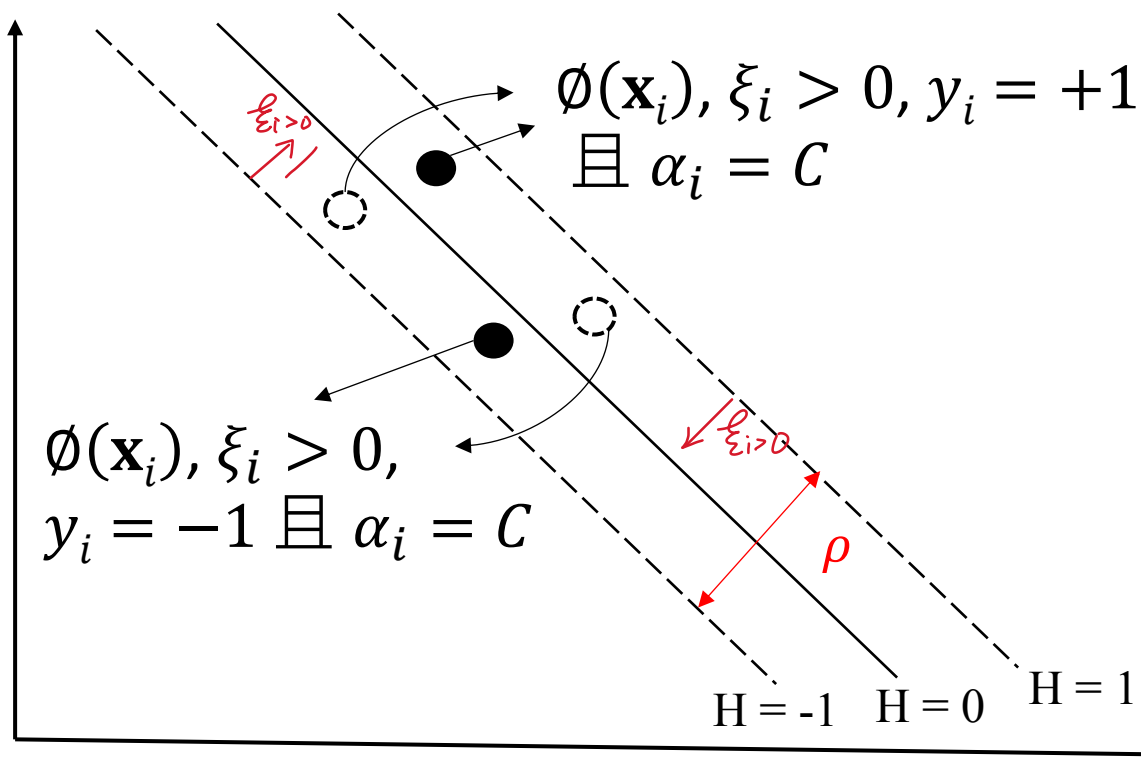
利用Kuhn-Tucker (KT) condition 求 b 值(Case 3)

Case 3 : 若 $\alpha_i = C \rightarrow \boxed{C - \alpha_i} = 0 \rightarrow \xi_i > 0$

KT conditions :

$$\alpha_i \times [y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i] = 0, \forall i$$

$$\beta_i \times \xi_i = 0, \forall i$$



又 $\alpha_i = C \neq 0$

$$\Rightarrow y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) - 1 + \xi_i = 0$$

(1) 若 $y_i = 1$, 則 $\mathbf{w}^T \phi(\mathbf{x}_i) + b = 1 - \xi_i$

(2) 若 $y_i = -1$, 則 $\mathbf{w}^T \phi(\mathbf{x}_i) + b = -1 + \xi_i$



核函數(Kernel Function)

Def :

If $K(,)$ is a function that satisfies Mercer's condition, then the inner product of two mapped data can be replaced by the kernel.

$$\rightarrow K(\mathbf{x}_i, \mathbf{x}_j) = K(\mathbf{x}_j, \mathbf{x}_i) \equiv \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$$

其中 $\langle , \rangle$: inner product operator · ϕ : nonlinear mapping

➤ Two commonly-used kernel functions :

① 多項式核函數(Polynomial kernel)

$$\rightarrow K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^P \text{ or } (\mathbf{x}_i^T \mathbf{x}_j)^P \equiv \phi(\mathbf{x}_i) \cdot \phi(\mathbf{x}_j)$$

⚠ No need to know exactly the nonlinear mapping ϕ , where P is the degree of the polynomial



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幅射基底函數(Radial Basis Function kernel , RBF)

② 幅射基底函數(Radial Basis Function kernel , RBF)

$$\begin{aligned}\rightarrow K(\mathbf{x}_i, \mathbf{x}_j) &= \phi(\mathbf{x}_i) \cdot \phi(\mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) = \phi(\mathbf{x}_j)^T \phi(\mathbf{x}_i) \\ &= e^{\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}}\end{aligned}$$

其中 σ : parameter of the RBF kernel , $\sigma \in [-\infty, \infty]$

Note

- ① 設定不同的核參數(kernel parameter)的值，等同於將data映射至不同的feature space中
setting different values of σ
 $\Leftrightarrow$ using a different nonlinear mapping function ϕ
 $\Leftrightarrow$ mapping the data $\mathbf{x}_i$ to different feature spaces F
- ② The best value of the kernel parameter is the one resulting in the highest classification accuracy
- ③ If we use a polynomial kernel and set $p = 1$, then we find the OSH in the original space(no mapping)
- ④ RBF kernel function = Gaussian kernel function



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高斯核函數的性質 (手算)



Summary of SVM

Training Phase

Given a training set $S = \{(\mathbf{x}_i, y_i)\}_{i=1,2,\dots,N}$, $\mathbf{x}_i \in \mathbb{R}^d$, y_i being either +1 or -1
Choose a kernel function and initialize the **kernel parameter** and **penalty weight C**

Step 1 : Solve the constrained optimization problem to obtain $\alpha_1, \alpha_2, \dots, \alpha_N$

$$\begin{aligned} &\text{Maximize} && \sum_i^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \\ &(\alpha_i \dots \alpha_N) && \\ &\text{subject to} && \sum_{i=1}^N \alpha_i y_i = 0 \\ &&& 0 \leq \alpha_i \leq C, \forall i = 1, \dots, N \end{aligned}$$



Summary of SVM

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Step 2 : Calculate the optimal value of bias b :

$$b^* = \frac{1}{y_k} - \sum_{i=1}^N \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}_k) \quad , \text{where } \mathbf{x}_k \text{ can be any training data whose } \alpha_k \text{ satisfies } 0 < \alpha_k < C$$

Test Phase (online stage)

For an unseen data(test data) $\mathbf{x} \in \mathbb{R}^d$, its class label is predicted by the SVM decision function:

$$D(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b^*$$

The decision rule is : If $D(\mathbf{x}) \geq 0$, then $\mathbf{x}$ is classified as positive class ; otherwise, $\mathbf{x} \in$ negative class.

Note

$$\because \mathbf{w} = \sum_{i=1}^N \alpha_i y_i \phi(\mathbf{x}_i) \quad \therefore D(\mathbf{x}) = \sum_{i=1}^N \alpha_i y_i \phi(\mathbf{x}_i)^T \phi(\mathbf{x}) + b^* = \sum_{i=1}^N \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b^*$$



Support Vector machine (SVM) 的命名由來

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S 中，某些 training data 的 α_i 值 $= 0$ ，others $\neq 0$ ($0 < \alpha_i < C, \alpha_i = C$)

$$\therefore D(\mathbf{x}) = \sum_{\mathbf{x}_j \in SV} \alpha_i y_i K(\mathbf{x}, \mathbf{x}_j) + b^*$$

$$SV \equiv \{\mathbf{x}_j | 0 < \alpha_i \leq C\}$$

, where the training data points $\mathbf{x}_j$ are called support vectors.

Solving this algorithm's dual problem is equivalent to **producing** a set of support vectors.

This is why this algorithm was called “Support Vector Machine (SVM)”.



SVM補充

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矩陣化(表示法)

Recall dual problem

$$\text{求 } \alpha_i \dots \alpha_N \begin{cases} \text{maximize} & \sum_i \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) \\ \text{subject to} & \sum_{i=1}^N \alpha_i y_i = 0 \quad 0 \leq \alpha_i \leq C, \forall i \end{cases}$$

若令 $\boldsymbol{\alpha} \in \mathbb{R}^N = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{bmatrix}$, Let $K(\mathbf{x}_i, \mathbf{x}_j) = K_{ij}$, 假設 $N=2$

$$\sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j) = \alpha_1 \alpha_1 y_1 y_1 K_{11} + \alpha_1 \alpha_2 y_1 y_2 K_{12} + \alpha_2 \alpha_1 y_2 y_1 K_{21} + \alpha_2 \alpha_2 y_2 y_2 K_{22}$$

$$= [\alpha_1 \ \alpha_2] \begin{bmatrix} y_1 y_1 K_{11} & y_1 y_2 K_{12} \\ y_2 y_1 K_{21} & y_2 y_2 K_{22} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

$$\text{又 } \sum_{i=1}^N \alpha_i = \alpha_1 + \alpha_2 + \dots + \alpha_N = \boldsymbol{\alpha}^T \mathbf{1} \rightarrow \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$\rightarrow \sum_{i=1}^N \alpha_i y_i = 0 \text{ 若令 } \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}, \text{ 叫做label vector } \rightarrow \mathbf{y}^T \boldsymbol{\alpha} = \boldsymbol{\alpha}^T \mathbf{y} = 0$$



矩陣化(表示法)

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因此Dual Problem的矩陣表示法如下：

$$\begin{cases} \text{maximize} & \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha} \\ \text{subject to} & \mathbf{y}^T \boldsymbol{\alpha} = 0 \\ & \mathbf{0} \leq \boldsymbol{\alpha} \leq \mathbf{C} \end{cases}$$

quadratic form

, where **K: kernel matrix** =

$$\begin{bmatrix} \alpha_1 \alpha_1 y_1 y_1 K_{11} & \cdots & \alpha_1 \alpha_N y_1 y_N K_{1N} \\ \vdots & \ddots & \vdots \\ \alpha_N \alpha_1 y_N y_1 K_{N1} & \cdots & \alpha_N \alpha_N y_N y_N K_{NN} \end{bmatrix}$$

N x N symmetric

Note :

$$\text{Maximize } \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha} \Leftrightarrow \text{Minimize } -\boldsymbol{\alpha}^T \mathbf{1} + \frac{1}{2} \boldsymbol{\alpha}^T \mathbf{K} \boldsymbol{\alpha}$$

總結：

SVM中，需要利用二次規劃(quadratic programming)方法來解Dual Problem



補充：二次式(quadratic form)

$$\text{若令 } \mathbf{x} = [x_1 \ x_2]^T, \text{ 若 } \mathbf{Q} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\begin{aligned} \text{則 } \mathbf{x}^T \mathbf{Q} \mathbf{x} &= [x_1 \ x_2] \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= [2x_1 + x_2 \quad x_1 + 3x_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1(2x_1 + x_2) + x_2(x_1 + 3x_2) \\ &= 2x_1^2 + x_1^1 x_2^1 + x_2^1 x_1^1 + 3x_2^2 \end{aligned}$$

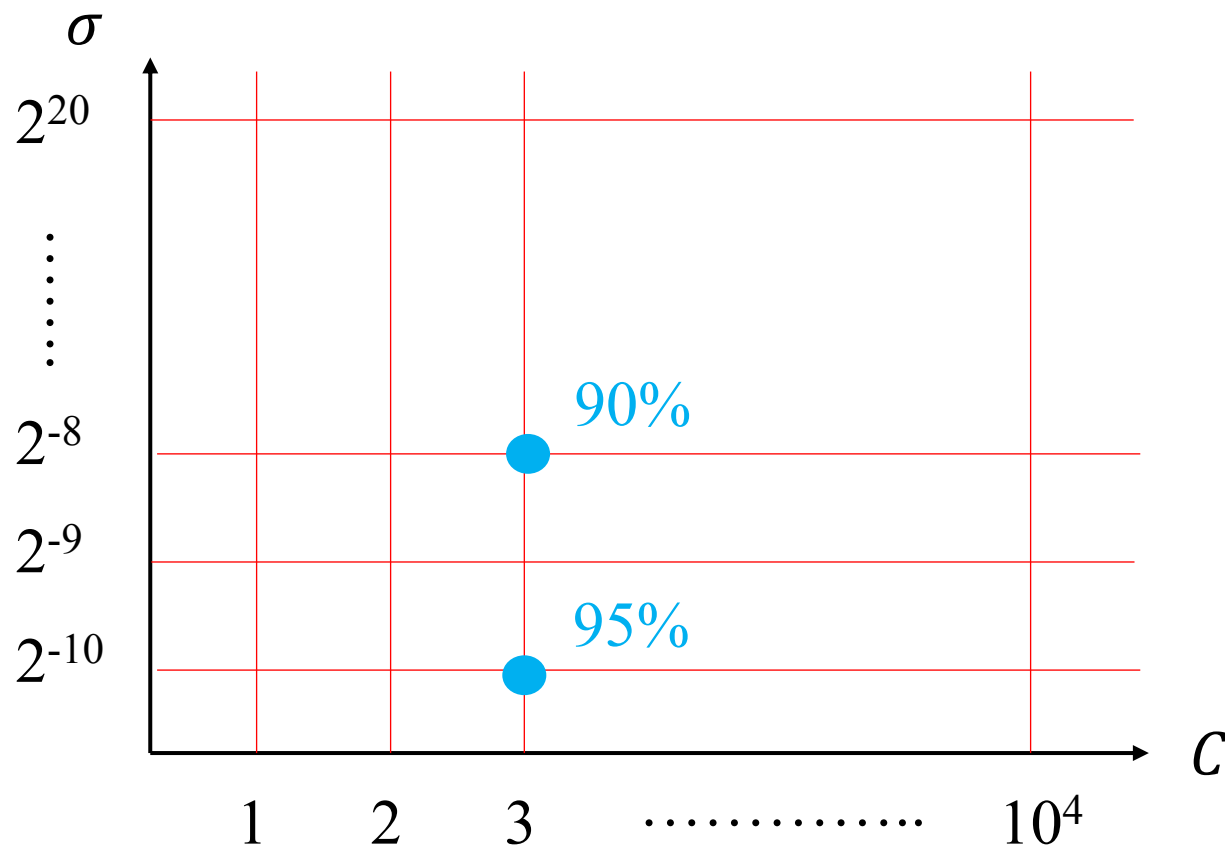
因此稱 $\mathbf{x}^T \mathbf{Q} \mathbf{x}$ 為二次式



SVM的參數最佳化(Hyperparameter Optimization)

定義：參數最佳化就是在找哪個參數組合能夠有最好的分類表現

格子搜尋法(Grid Search)



$(C = 3, \sigma = 2^{-10})$ 的參數組合得到的分類率比 $(C = 3, \sigma = 2^{-8})$



SVM的參數最佳化 (Hyperparameter Optimization)

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總結：結合格子搜尋法與交叉驗證來決定最佳參數

給定一個訓練集合 $S = \{(\mathbf{x}_i, y_i)\}$, $i = 1, \dots, N$, 先初始化 C, σ
(以RBF kernel的參數為例)

Step 1：先對訓練集合做 K 摺交叉驗證

Step 2：記錄 K 摺交叉驗證的分類率及參數組合 (C, σ)

Step 3：
$$\begin{cases} C \leftarrow C + \Delta C, \text{ 重複Step 1和2直到所有的參數組合都已測試過} \\ \sigma \leftarrow \sigma + \Delta \sigma \end{cases}$$

Step 4：
$$(C^*, \sigma^*) = \arg \max_{(C, \sigma)} CR(C, \sigma)$$



ML classification methods

1. K-nearest neighbor (K-NN) classifier
2. Linear discriminant analysis (LDA) classifier
3. Quadratic discriminant analysis (QDA)
4. Mahalanobis Discriminant Analysis (MDA) classifier
5. Support vector machine (SVM) : a binary classifier
6. Multi-class SVM classifier

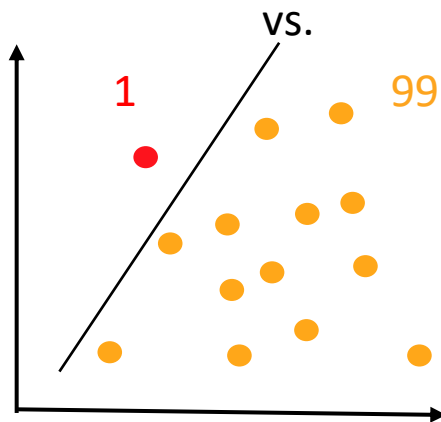


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SVM的多類別分類(multi-class SVM classification)

Most common methods :

- ① One-Against-One(OAO) method 一對一方法
- ② One-Against-All(OAA) or One versus rest
 - ➔可能產生不平衡的資料集(imbalanced dataset)
 - ➔會讓結果產生偏差(biased decision boundary)



➔ 為了讓大類別判對，導致偏向小類別



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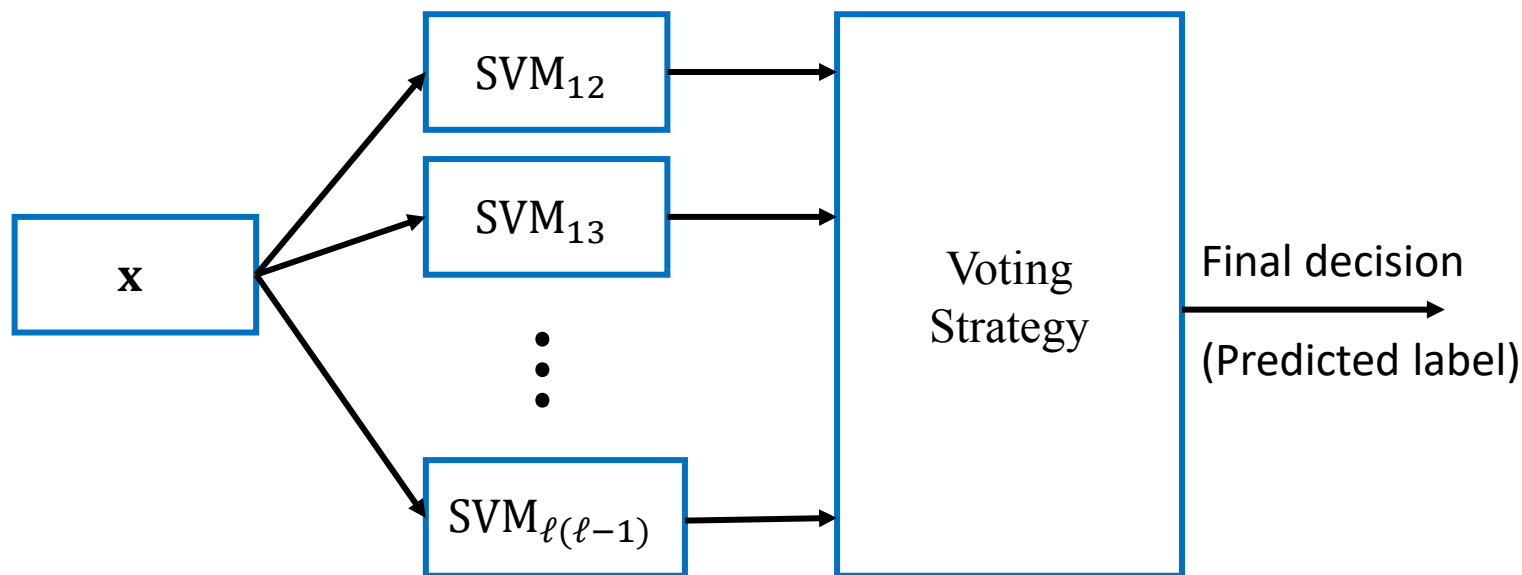
One-Against-ALL(OAA) method 手抄範例



One-Against-One(OAO) method 一對一方法

Assume : ℓ 個 classes to be classified

ℓ 個 classes 當中，任意兩個class訓練出一個SVM，共訓練出 $\frac{\ell(\ell-1)}{2}$ 個SVM。若 $\mathbf{x}$ 為一筆test data，將 $\mathbf{x}$ 輸入至這些訓練好的SVM，得到 $\frac{\ell(\ell-1)}{2}$ 個 predicted class labels，最後再利用投票策略(Voting Strategy)決定最終的決定(final decision)



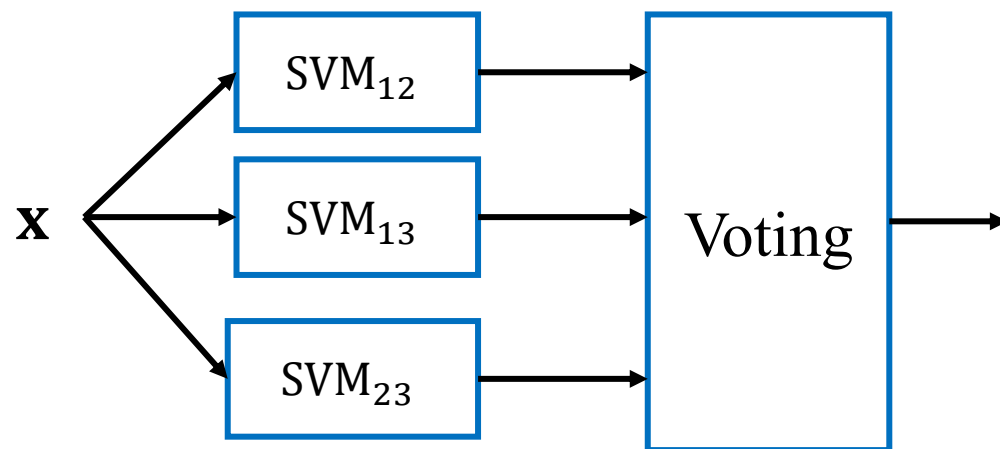


以Iris data 為例

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$$\ell = 3$$

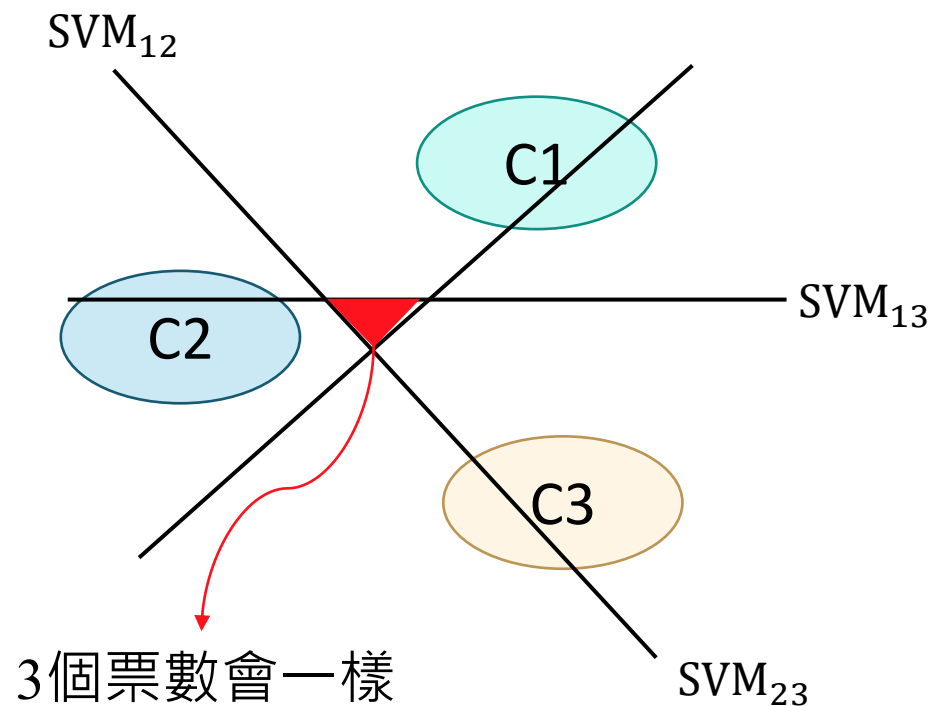
- ① Train 3個SVM，分別用來classify
- | | | |
|---|-----------|------------------------|
| { | C1 vs. C2 | 共3個 binary classifiers |
| | C1 vs. C3 | |
| | C2 vs. C3 | |
- ② Testing phase : (if $\mathbf{x}$ is a test data)





◆ 如果票數一樣?

◆ Other issues?



3個票數會一樣

➔ unclassified

➔ **classification error**